Kanageshwaran Dhakshinamoorthy

Stat 50

HW 22
Section 6.9

1

(a) $H_0: P_1 = 0.85$, $P_2 = 0.10$, $P_3 = 0.05$

vs H_1 : any one on H_0 is not true. $n = 500$

(b) $\chi^2 = \sum_{i=1}^{3}$ ~~$(O_i - E_i)$~~

	Freq O_i	E_i
Conforming	405	$500(.85) = 425$
downgraded	55	$500(.10) = 50$
Scrap	40	$500(.05) = 25$

(c) $\chi^2 = \sum_{i=1}^{3} \frac{(O_i - E_i)^2}{E_i}$

$= \frac{(405-425)^2}{425} + \frac{(55-50)^2}{50} + \frac{(40-25)^2}{25}$

$= 10.44117647$

(d) $0.005 < P < 0.01$ $(P = 0.005404)$

Fail to Support. There is enought evidence at commonly use $\alpha = 0.1, 0.05, 0.$ to conclude that the true percentage differs from 85%, 10%, and 5%

Supplement 18-

5-/. Ch

① Number of sixes	O_i	E_i
0	480	$(.58)(1000) = 580$
1	350	$(.345)(1000) = 345$
2	140	$(.07)(1000) = 70$
3	030	$(.005)(1000) = 5$

H_0: $P_1 = 0.58$, $P_2 = 0.345$, $P_3 = 0.07$, $P_4 = 0.005$

① $E_0 \Rightarrow 580$
$E_1 \Rightarrow 345$
$E_2 \Rightarrow 70$
$E_3 \Rightarrow 5$

$n = 4$
$d.f = 3$

② H_1: $P_1 \neq 0.58$ or $P_2 \neq 0.345$ or $P_3 \neq 0.07$ or $P_4 \neq 0.005$

④ $$\chi^2 = \sum_{i=1}^{4} \frac{(O_i - E_i)^2}{E_i}$$

$$= \frac{(480-580)^2}{580} + \frac{(350-345)^2}{345} + \frac{(140-70)^2}{70} + \frac{(30-5)^2}{5}$$

$$= 212.3138431$$

⑤ $(P < 0.005)$

Fail to support the claim. There are a strong evidence to fail the claim that dices are fair

at the 5% Significance level.

Supplement 2:-

$\lambda = 10$

$n = 100$

Number of Hits	Count	Expected # Hits
8	8	10
9	11	10
12	15	10
14	1	10

5% Poisson(10)

H_0: $P_1 = 0.06306$, $P_2 = 0.12511$, $P_3 = 0.094780$, $P_4 = 0.85207$

H_1: any one of the H_0 is not true.

$$\chi^2 = \sum_{i=1}^{4} \frac{(O_i - E_i)^2}{E_i}$$